Ashish Gupta

Senior Software Engineer | Distributed Systems & Applied AI

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Senior Software Engineer with **9+ years of experience** architecting enterprise distributed systems, high-performance web applications, and mission-critical microservices across Testing, BFSI, Supply Chain, and Energy sectors. Expert in full-stack engineering with **C#, .NET 8, TypeScript, and SQL**, alongside production systems delivery in **Rust and Java**. Proven track record building event-driven pub/sub pipelines, 100+ GB high-precision data reconciliation engines, and production AI integrations including custom Model Context Protocols (MCPs). Living in Vienna with unrestricted Austrian labor market mobility under **Rot-Weiß-Rot-Karte Plus (RWR+)**.

SKILLS & TECHNOLOGIES

- **Languages & Runtimes:** C#, TypeScript, Rust, SQL (T-SQL, PostgreSQL), JavaScript, Java, Python.
- **Backend & Distributed Systems:** .NET 8, ASP.NET Core, Entity Framework (EF Core), Web APIs, gRPC, SignalR (WebSockets), Azure Service Bus (pub/sub), Event-Driven Architecture, Microservices.
- **AI & Agentic Systems:** Model Context Protocol (MCP) servers (JSON-RPC, stdio & HTTP), LLM Tool-Calling APIs, Natural Language-to-SQL (NL-to-SQL), Automated Schema Discovery.
- **Cloud, DevOps & Observability:** Azure Cloud, Docker, Kubernetes, Helm Charts, Azure DevOps CI/CD, GitHub Actions, OpenTelemetry (Distributed Tracing), Infrastructure as Code (NUKE), Snyk.
- **Frontend & Tooling:** React, Angular, RxJS, Next.js, Monaco Editor, Vite, Nx Monorepo, AG-Grid, Tailwind CSS, Material UI (MUI).
- **Databases & Data Formats:** SQL Server, Fluent Migrator, MongoDB, SQLite, Apache Parquet, Avro, Apache Hive, Azure Blob Storage (100+ GB high-precision reconciliation).
- **Testing & Quality Engineering:** Test-Driven Development (TDD), Behavior-Driven Development (BDD), SpecFlow, xUnit, Unit & Integration Testing, End-to-End Testing, Jest, Playwright.

PROFESSIONAL EXPERIENCE

Tricentis — **Senior Software Engineer** Vienna, Austria | **June 2024 – Present**

- **AI-Powered Product Integration & MCPs:** Engineered and delivered custom Model Context Protocols (MCPs) bridging LLMs with hybrid cloud and on-premise test environments, establishing automated schema discovery, Natural Language-to-SQL synthesis, and automated test-case generation to slash authoring friction for enterprise clients.
- **Decoupled Event-Driven Messaging & Real-Time Sync:** Architected an enterprise pub/sub messaging pipeline using Azure Service Bus, a dedicated Notification Service, and SignalR (WebSockets), propagating state updates and automated MCP tool calls to web clients with debounced re-fetch and live UI refresh.
- **100+ GB Data Reconciliation Engine:** Maintained and optimized the Tosca DI row-by-row comparison engine, reconciling 100+ GB datasets with high precision across SQL databases, Apache Hive, and heterogeneous file formats (CSV, Excel, Parquet, Avro).
- **Enterprise Test Intelligence Dashboard:** Built a GitHub Actions post-execution reporting pipeline and interactive dashboard for 1,396+ nightly regression test runs, scaling adoption across 19 engineering teams as the primary daily standup failure-triage platform.
- **High-Performance Desktop Tooling:** Maintained and enhanced a mission-critical desktop reporting tool, utilizing Rust for low-latency, memory-safe data processing paired with an Electron frontend.
- **Interactive SQL Playground & UI Tooling:** Engineered a web-based SQL playground micro-frontend using Monaco Editor, enabling users to format and execute live queries with real-time syntax validation, IntelliSense, and low-latency result rendering.

International Atomic Energy Agency (IAEA, via e-Zest) — **Senior Software Engineer** Vienna, Austria | **April 2023 – May 2024**

- **Mission-Critical Safeguards Tooling:** Architected and enhanced core components of the SAFIRE reporting application (.NET Core, Angular, SQL Server) and integrated auxiliary inspection systems, ensuring high availability for international nuclear inspectors worldwide.
- **Quality Engineering & Verification:** Applied TDD and BDD methodologies to deliver 10+ core inspection features, strengthening automated verification frameworks and reducing data validation errors by 25%.
- **Stakeholder & Inspector Workflow:** Partnered directly with nuclear inspectors and domain stakeholders to translate inspection workflows into automated reporting features and seamless auxiliary integrations.

LTIMindtree — **Module Lead** Mumbai, India | **April 2021 – April 2023**

- **Compliance Engineering:** Engineered a full-stack Remote Worker Assessment web application for Deloitte (U.S.) using .NET and Angular, guaranteeing 100% regulatory compliance across 15+ international labor laws.

- **System Architecture & Decoupling:** Drove microservices decoupling, secure RESTful API development, and database query optimization, reducing latency across core assessment pipelines.

GEP Worldwide — **Senior Software Engineer (Promoted in 12 mos)** Mumbai, India | **September 2019 – April 2021**

- **Logistics Visualization & Telematics:** Integrated Azure Maps with the enterprise Control Tower UI, delivering real-time shipment tracking that reduced client inquiry queries by 25%.
- **Reusable UI Frameworks & Analytics:** Architected reusable dashboard components across Demand Planning projects, enabling high-performance visualization for complex historical and forecasted datasets.
- **Data Strategy & Ingestion:** Designed high-performance data ingestion pipelines and visualization mechanisms utilizing NoSQL database architectures for high-volume supply chain transactions.

TSS Consultancy — **Jr Software Engineer** Mumbai, India | **April 2018 – September 2019**

- **Cryptography & Financial Security:** Built banking security modules utilizing Public/Private key cryptography and digital certificates across Java and .NET environments, eliminating data-in-transit vulnerabilities.
- **CRM Database Consolidation:** Architected a consolidated customer relationship management application unifying fragmented datasets into a single, high-integrity transactional database.

AUM Insurance Brokers — **Software Engineer** Mumbai, India | **June 2017 – April 2018**

- **E-Commerce & API Integration:** Built TakeAPolicy.com, a public-facing insurance portal integrating vendor REST APIs for real-time policy comparisons, automated premium calculations, and secure payments.

EDUCATION

Bachelor of Computer Science — University of Mumbai Mumbai, India | 06/2012 – 05/2016

LANGUAGES & LEGAL WORK AUTHORIZATION

- **Languages:** English (Full Professional Proficiency, C1/C2), German (A1/A2), Hindi (Native).
- **Work Authorization:** Rot-Weiß-Rot-Karte Plus (RWR+) — Unrestricted Austrian labor market mobility with full employer portability and zero sponsorship requirements.

AWARDS & ACCELERATED ADVANCEMENT

- **Rapid Advancement at Tricentis:** Formally promoted to Senior Software Engineer in January 2026 within 19 months of joining, recognizing impactful engineering contributions across cloud microservices, desktop tooling, and AI agent initiatives.
- **Accelerated Promotion at GEP Worldwide:** Formally promoted from Software Engineer to Senior Software Engineer within 12 months of tenure (Q4-2020) for driving delivery on critical supply chain visibility tools.